

Fahad Bin Aziz Nabil

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PROFESSIONAL SUMMARY

Backend developer with hands-on experience building scalable applications using **ASP.NET Core**, **Entity Framework Core**, and **SQL Server**. Skilled in designing **RESTful APIs**, implementing authentication and authorization, and applying **Clean Architecture** principles. Built **microservices** using **YARP**, **RabbitMQ**, **Apache Kafka**, **gRPC**, **Redis**, and **Docker** through academic and personal projects. Strong problem-solving skills demonstrated through competitive programming on **Codeforces**. Passionate about building reliable, maintainable, and high-performance backend systems, with a strong interest in expanding expertise in **DevOps**, cloud infrastructure, and distributed systems.

EDUCATION

Chittagong University of Engineering and Technology (CUET) <i>B.Sc. in Computer Science and Engineering</i>	Mar 2022 – Aug 2026 CGPA: 3.60 / 4.00
Notre Dame College, Dhaka <i>Higher Secondary Certificate (Science)</i>	Jun 2018 – Dec 2020

TECHNICAL SKILLS

Programming Languages: C, C++, C#, Python, SQL.

Backend Development: ASP.NET Core (MVC, Web API), Entity Framework Core, ASP.NET Core Identity, SignalR, MailKit, REST, gRPC.

Databases & Messaging: Microsoft SQL Server, PostgreSQL, MongoDB, MySQL, SQLite, RabbitMQ, Apache Kafka.

Architecture, Security & DevOps: Microservices, Clean Architecture, Layered Architecture, Repository Pattern, Event-Driven Architecture, YARP API Gateway, Redis, JWT, OAuth2, OpenID Connect, Docker, Docker Compose.

Testing & Tools: xUnit, Git, GitHub, Swagger / OpenAPI, Postman, Visual Studio, VS Code.

Machine Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, Pandas, NumPy, CNNs, Vision Transformers.

PROJECTS

Residential Complex Manager | *Ongoing*

- Architecting a service-per-domain **microservices** platform using **YARP** as the API Gateway and containerizing services with **Docker Compose**.
- Implementing authentication and authorization using **OAuth2**, **OpenID Connect (OIDC)**, and **JWT**, while integrating **PostgreSQL**, **MongoDB**, and **SQL Server** based on service-specific data requirements.
- Implementing asynchronous event-driven communication using **RabbitMQ** and **Apache Kafka**, with **gRPC** for low-latency synchronous service-to-service communication.
- Integrating **Redis** for distributed caching and developing comprehensive **xUnit** unit tests to improve application performance and validate business logic.
- Stack:* .NET 8, ASP.NET Core, YARP, OpenAPI, gRPC, RabbitMQ, Apache Kafka, Redis, PostgreSQL, MongoDB, SQL Server, SQLite, Docker, xUnit.

Student Hall Management System — Backend API

- Developed a layered **ASP.NET Core Web API** following Controllers, Services, Repositories, DTOs, Middleware, and Helpers architecture, exposing **50+ RESTful API endpoints** for room

allotment, student registration, payment tracking, complaint management, and administrative dashboards across **three user roles** (DSW Admin, Hall Admin, Student).

- Implemented **JWT**-based authentication and role-based authorization, along with centralized password hashing and CORS configuration to secure API access.
- Documented the complete API using **Swagger** / **OpenAPI**, validated endpoints with **Postman**, and integrated SMTP email notifications for account registration and complaint workflows.
- *Stack:* .NET 8, ASP.NET Core Web API, Entity Framework Core, SQL Server, JWT, Swagger / OpenAPI, Postman.

News Portal — Role-Based News Publishing Platform

- Developed a role-based news management platform with **three user roles** (Admin, Author, User) following **Clean Architecture** across separate UI, Core, and Infrastructure projects, comprising **5 controllers**, **30+ RESTful endpoints**, and **15+ MVC pages**.
- Implemented authentication and authorization using **ASP.NET Core Identity**, including **Google OAuth**, email confirmation, and password reset via **MailKit** SMTP integration.
- Built editorial workflows for article publishing, moderation, content pinning, and role-based Areas, while integrating **ImageSharp** for image processing and optional **Cloudinary** storage.
- *Stack:* .NET 8, ASP.NET Core MVC, Entity Framework Core, SQL Server, ASP.NET Core Identity, MailKit, ImageSharp, Cloudinary, Google OAuth.

Used Car Price Prediction Engine

- Developed a **stacking ensemble** comprising **XGBoost**, **LightGBM**, and **CatBoost** base learners with a meta-learner, optimized using **Optuna** and loaded from **12 serialized model artifacts** for inference.
- Engineered a 12-feature preprocessing pipeline using **Pandas**, **NumPy**, **scikit-learn**, and **category_encoders**, incorporating structured vehicle attributes and TF-IDF text features.
- Integrated the trained Python model with an **ASP.NET Core MVC** application by executing a dedicated Python inference process, enabling seamless predictions through the web interface; containerized the complete solution using a multi-stage **Dockerfile** and deployed it on Render.
- *Stack:* Python 3, scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, .NET 8, ASP.NET Core MVC, Docker.

Lost and Found Web Portal — Backend API

Industrial Attachment Project at Synesis IT PLC.

- Developed the complete backend using **ASP.NET Core**, **Entity Framework Core**, and **SQL Server**, exposing **10+ RESTful API endpoints** for item reporting, user management, and lost-and-found matching workflows.
- Implemented **JWT**-based authentication with refresh-token rotation and integrated **SignalR** for real-time messaging between item owners and finders.
- Documented APIs using **Swagger** / **OpenAPI** to streamline client integration and deployed the backend alongside a separate frontend application.
- *Stack:* .NET 8, ASP.NET Core Web API, Entity Framework Core, SQL Server, JWT, SignalR, Swagger / OpenAPI.

EXTRACURRICULAR

Competitive Programming

Solved **600+** algorithmic problems on [Codeforces](#) (handle: **DNZLREV**), achieving a peak rating of **1507 (Specialist)**, with additional practice on other online judges. Strengthened problem-solving skills in data structures and algorithms.

Vice President — Greater Kushtia Association, CUET

- Coordinated cultural, social, and professional events for the regional student community at CUET.
- Managed a **30+ member** executive committee and collaborated with university administration to organize events and oversee logistics.